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ADAPTIVE DIGITAL SAT PRACTICE TEST

Horizon Assessment Series — Form A

Complete Adaptive Format with Easier and Harder Module 2 Paths

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- **Section 1: Reading and Writing** (81 Questions Total)
- **Section 2: Math** (66 Questions Total)
- Separate Answer Key & Detailed Explanations

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QUICK-REFERENCE ANSWER GRIDS

Section 1 — Module 1 (27 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	2	B	3	A	4	A	5	C
6	B	7	D	8	**C**	9	B	10	A
11	C	12	A	13	**C**	14	D	15	A
16	B	17	D	18	D	19	A	20	**C**
21	D	22	B	23	**A**	24	B	25	C
26	B	27	D						

Distribution: A=7, B=7, C=6, D=7

Section 1 — Module 2 Easier (27 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	D	3	B	4	C	5	A
6	B	7	C	8	B	9	A	10	D
11	B	12	A	13	C	14	D	15	B
16	D	17	D	18	C	19	B	20	C
21	D	22	D	23	**A**	24	A	25	A
26	A	27	B						

Distribution: A=7, B=7, C=6, D=7

Section 1 — Module 2 Harder (27 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	2	A	3	B	4	**D**	5	A
6	**C**	7	B	8	A	9	C	10	**D**
11	B	12	B	13	**C**	14	C	15	D
16	**D**	17	D	18	B	19	B	20	C
21	A	22	D	23	A	24	**C**	25	A
26	A	27	B						

Distribution: A=7, B=7, C=6, D=7

Section 2 — Module 1 (22 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	2	A	3	A	4	9	5	D
6	A	7	**C**	8	D	9	13	10	C
11	B	12	B	13	B	14	44	15	300
16	C	17	42	18	**C**	19	A	20	C
21	D	22	B						

MC Distribution: A=4, B=4, C=5, D=4

Section 2 — Module 2 Easier (22 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	3250	3	B	4	A	5	A
6	C	7	C	8	**D**	9	22.4	10	B
11	C	12	B	13	**D**	14	**C**	15	A
16	B	17	D	18	D	19	C	20	80
21	−18	22	2187						

MC Distribution: A=4, B=4, C=5, D=4

Section 2 — Module 2 Harder (22 Qs)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	**A**	2	75	3	**A**	4	D	5	250
6	C	7	C	8	**D**	9	A	10	55
11	**A**	12	B	13	66	14	−36	15	4096
16	B	17	D	18	C	19	B	20	B
21	7	22	C						

MC Distribution: A=4, B=4, C=4, D=3

SECTION 1: READING AND WRITING

MODULE 1 (Same for all students — 27 Questions)

Question 1

Skill: Words in Context | Difficulty: Medium

In several of her textile installations, Sheila Hicks employs vibrant chromatic threads to construct her subjects rather than rendering them in muted tones. For example, her 2017 piece *Escalade Beyond Chromatic Lands* is intensely colorful and therefore stands apart from some of her other works in which the viewer can easily ______ subdued hues.

Which choice completes the text with the most logical and precise word or phrase?

A) appreciate
B) overlook
C) fabricate
D) discern

Question 2

Skill: Words in Context | Difficulty: Medium

During the early decades of commercial radio, those who owned stations had a financial interest in persuading listeners to tune in regularly. As media historian Susan Douglas argues, some broadcasters relied on dramatic programming to shape the public's ______ the new medium. Programs that emphasized storytelling techniques borrowed from established literary forms, like serialized fiction, could help legitimize radio as a cultural force.

Which choice completes the text with the most logical and precise word or phrase?

A) contribution to
B) perception of
C) reproduction of
D) application to

Question 3

Skill: Words in Context | Difficulty: Medium

The butterfly exhibits at conservatories such as the Pacific Science Center in Seattle (which houses a tropical butterfly enclosure among its attractions) are notable for the ______ of the preparation behind them—the staff consulted entomologists and tropical ecologists extensively to ensure the authenticity of the habitat conditions.

Which choice completes the text with the most logical and precise word or phrase?

A) rigor
B) obscurity
C) shallowness
D) novelty

Question 4

Skill: Words in Context | Difficulty: Hard

Technology clusters emerge when numerous firms in complementary sectors ______ a region, as with semiconductor designers and fabrication facilities in the greater Taipei area. Analysts have long assumed that firms concentrate for identical reasons, but Maryann Feldman and colleagues found that the factors propelling clustering in certain cases are only loosely associated with clustering in others.

Which choice completes the text with the most logical and precise word or phrase?

A) amass in
B) appeal to
C) intercede in
D) concur with

Question 5

Skill: Words in Context | Difficulty: Medium

Contemporary concert halls are routinely outfitted with sophisticated acoustic panels and digital sound systems to optimize the audience’s listening experience. Because concertgoers have grown accustomed to these carefully engineered environments, performances in raw, untreated spaces can startle audiences. But plain acoustic settings were likely ______ audiences in much earlier periods: purpose-built concert halls with advanced acoustics were not widespread until the 1800s.

Which choice completes the text with the most logical and precise word or phrase?

A) disliked by
B) confusing to
C) expected by
D) exciting to

Question 6

Skill: Main Purpose | Difficulty: Medium

Founded in 2003 in Detroit, Michigan, the Heidelberg Project is an outdoor art installation that transforms vacant lots and abandoned houses on the city's east side into large-scale works of art. Created by artist Tyree Guyton, the project uses found objects—old shoes, car hoods, stuffed animals—to cover entire building facades, turning symbols of urban decline into statements about community resilience. While some neighbors initially objected to the unconventional aesthetics, the project has attracted international attention and is credited with reducing illegal dumping and arson in the surrounding blocks.

Which choice best states the main purpose of the text?

A) To argue that outdoor art installations are more effective than traditional urban renewal programs
B) To describe a project that repurposes signs of neglect into art while affecting the surrounding community
C) To explain why residents of Detroit's east side have historically opposed public art initiatives
D) To compare the Heidelberg Project with other community art programs across the United States

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Question 7

Skill: Overall Structure | Difficulty: Hard

The following text is adapted from a 1905 novella by Edith Wharton. Mrs. Brympton is the mistress of a country estate.

By the servants of the estate and its surrounding parish, Mrs. Brympton was regarded as a woman of great generosity and impeccable manners, who hosted charitable functions with evident grace and never failed to inquire after the welfare of the tenants' families.
Mrs. Brympton never initiated a disagreeable exchange; she listened to what others proposed,

but seldom expressed what she herself desired; she never permitted anyone to perceive that she was weary of their company, and she never afforded any person a reason for speaking ill of her.

Which choice best describes the overall structure of the text?

A) It shows that Mrs. Brympton had originally been held in low esteem by her neighbors and then explains how her charitable acts gradually improved their opinion of her.
B) It highlights the contrast between Mrs. Brympton's public generosity and her private indifference and then explains why her servants ultimately grew suspicious of her.
C) It implies that Mrs. Brympton's neighbors are more discerning in their judgment of her than her servants are and then explains why her servants were so easily deceived.
D) It presents the favorable opinion of Mrs. Brympton that others hold and then describes the behaviors of Mrs. Brympton that enable her to maintain that favorable opinion.

Question 8

Skill: Overall Structure | Difficulty: Hard

Lucia Martinez and colleagues have studied how bioluminescence—the capacity of an organism to produce light through chemical reactions—evolved independently in multiple deep-sea lineages. It had been widely accepted that bioluminescence arose through a shared ancestral gene. Meanwhile, Yuri Oba and colleagues have examined how different organisms achieve luminescence via distinct biochemical pathways, but the relative frequency with which these pathways emerge through shared versus independent genetic changes remains poorly mapped. This motivated marine biologists Takashi Kondo and Rina Fujimoto to catalogue both types of luminescence origins in a single comparative study for their 2018 publication.

Which choice best states the overall structure of the text?

A) It explains a widespread assumption about a phenomenon, describes two studies that are based on that assumption, and explains why the assumption is unfounded.
B) It discusses a particular study of a phenomenon, elaborates on how two other studies have approached this phenomenon in the past, and critiques the first study's approach.
C) It describes a phenomenon, provides examples of two studies of that phenomenon, and mentions a third study of that phenomenon.
D) It details the development of a field of scientific inquiry, gives examples of three studies that were important to this development, and speculates on how future studies can develop these ideas.

Question 9

Skill: Main Idea | Difficulty: Medium

The following text is from a translation of Natsume Soseki's 1914 novel *Kokoro*. The narrator, a young student, has just arrived in a coastal town for a summer holiday.

Through the course of my life there have been countless junctures when circumstances demanded that I act decisively, abrupt crossroads and unforeseen detours that I have had to navigate as best I could. Sometimes I rose to meet them; more often I stumbled. Yet never was I so keenly conscious of standing at a threshold as I was on that August afternoon when I first descended the stone steps toward the shore. What lay behind me was a tangle of incomplete obligations, and before me stretched an expanse—both oceanic and existential—whose vastness promised nothing except the chance to begin again.

Which choice best states the main idea of the text?

A) The narrator feels optimistic about the potential success of a new business opportunity.
B) The narrator recognizes that a particular moment represents a significant turning point in his life.
C) The narrator has always had a systematic approach for managing life's major transitions.
D) The narrator wishes to express gratitude toward the mentors who guided him to this destination.

Question 10

Skill: Inference / Complete the Text | Difficulty: Hard

In India, reliance on groundwater for irrigation as a share of total agricultural water use grew by approximately 40 percent between 1990 and 2015; such increases are commonly explained by reference to the productivity hypothesis, a framework holding that irrigation method is driven primarily by farm output (specifically, groundwater replaces surface water as yields rise). Priya Nair and Arvind Subramanian's analysis of groundwater use in Tamil Nadu shows this framework to be incomplete, however: household irrigation practices were variable, context-dependent, and shaped by multiple considerations, including regional rainfall patterns and the cost of electric pumping.

Based on the text, how would an advocate of the productivity hypothesis most likely explain the change observed in India?

A) Between 1990 and 2015, farm yields in India increased, leading farmers to meet a larger share of their irrigation needs with groundwater.
B) Beginning in 1990, groundwater extraction technology became more widely accessible in India, enabling farms of various sizes to shift away from surface irrigation.
C) Groundwater use expanded in India between 1990 and 2015 because fluctuations in rainfall during that period made surface water supplies less reliable relative to demand.
D) Farm yields in India rose from 1990 to 2015, which weakened the link between productivity and irrigation source.

Question 11

Skill: Evidence - Data / Graph / Table | Difficulty: Easy

GRAPH TO INSERT

Billions of Kilograms of Nickel Ore Extracted in 2000 and 2022

Country	2000	2022
Brazil	1.10	3.45
Australia	2.50	3.20
Chile	4.60	5.70
Indonesia	1.05	1.80

While preparing a report on global mining trends, a student examines nickel ore extraction across several countries in 2000 and 2022. The student observes that Chile extracted 4.60 billion kilograms of nickel ore in 2000 and _____

Which choice most effectively uses data from the table to complete the statement?

A) 3.45 billion kilograms in 2022.
B) 1.80 billion kilograms in 2022.
C) 5.70 billion kilograms in 2022.
D) 3.20 billion kilograms in 2022.

Question 12

Skill: Quotation Illustrates Claim | Difficulty: Hard

"Winter Memories" is an 1866 poem by Henry David Thoreau. In the poem, the speaker reflects on how the natural landscape in winter carries traces of the past, suggesting that _____

Which choice most effectively uses a quotation from "Winter Memories" to illustrate the claim?

A) a returning traveler notices that "the old familiar haunts / Have kept their winter secrecy, / And whisper still of former days / In language only known to me."
B) the frozen ground "reflects the silence of the sky, / A mirror of the season's calm, / Where nothing stirs beneath the frost."
C) the speaker "walks beneath the laden boughs / And counts each icicle in turn, / Admiring winter's craftsmanship."
D) the landscape "lies in perfect sleep / While travellers upon the road / Take note of every frozen stream."

Question 13

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

[Graph: A grouped bar chart titled "Listener Satisfaction: Familiar vs. Unfamiliar Music." The x-axis lists six musical pieces by title: "Nocturne," "March," "Rondo," "Fugue," "Waltz," and "Prelude." The y-axis shows average satisfaction rating from 0 to 10. For each piece, there are two bars—one light (unfamiliar) and one dark (familiar). For most pieces, the familiar bar is taller than the unfamiliar bar. However, the gap varies: for "Rondo" and "Prelude," the bars are nearly equal in height (both around 6.5); for "March" and "Waltz," the familiar bar is substantially taller (familiar ~7.5, unfamiliar ~4.5).]

Researchers examined how prior familiarity with a musical composition affects listener satisfaction. Participants rated their enjoyment of one piece they had heard before and one they had not. For every piece, those who were already familiar with the composition reported higher satisfaction than those who were not. But the magnitude of this difference varied across compositions, as is best illustrated by the satisfaction ratings for ______

Which choice most effectively uses data from the graph to complete the statement?

A) "Nocturne" and "Fugue."
B) "Rondo" and "Prelude."
C) "March" and "Waltz."
D) "Fugue" and "Prelude."

Question 14

Skill: Evidence - Support / Weaken | Difficulty: Hard

Although Atlantic bluefin tuna generally travel between spawning grounds in the Gulf of Mexico and feeding areas in the North Atlantic, a subset of this species—identified as the Eastern Atlantic stock—feeds along the coasts of Portugal and Morocco instead. Interestingly, individuals in this subset reach smaller maximum lengths than other Atlantic bluefin despite showing comparable juvenile growth rates. Marine biologists hypothesize that this size difference may be an adaptation to the particular prey availability in the Eastern Atlantic feeding range.

Which finding, if true, would most directly support the researchers' claim regarding the size of Eastern Atlantic stock tuna?

A) When feeding along the coasts of Portugal and Morocco, Eastern Atlantic stock tuna are in closer proximity to commercial fishing fleets than North Atlantic tuna are when feeding in open waters.

B) The average body length of Eastern Atlantic stock tuna observed along the coasts of Portugal and Morocco has remained relatively stable in recent decades, while the average body length of North Atlantic tuna has slightly decreased.
C) When present along the coasts of Portugal and Morocco, Eastern Atlantic stock tuna tend to forage in shallow coastal shelf areas inaccessible to tuna of the main North Atlantic group.
D) Certain small pelagic fish species available along the coasts of Portugal and Morocco where Eastern Atlantic stock tuna feed are not found in the open North Atlantic waters where the main group forages.

Question 15

Skill: Inference / Complete the Text | Difficulty: Hard

Chimamanda Ngozi Adichie's 2013 novel *Americanah* is frequently categorized as immigrant fiction. That label has some merit—there are clear thematic parallels between the novel's protagonist Ifemelu and Adichie's own experience of moving from Nigeria to the United States—but it should not be taken to mean that every character and event depicted in *Americanah* corresponds to real life. The novel is substantially a work of imagination, and readers who disregard this fact and instead attempt to map increasingly specific autobiographical parallels risk ______

Which choice most logically completes the text?

A) imposing unwarranted correspondences between *Americanah* and Adichie's life.
B) minimizing the degree to which Adichie drew on personal material when writing *Americanah*.
C) mischaracterizing *Americanah* as having a wider readership than it actually does.
D) overemphasizing the extent to which Adichie borrowed techniques from earlier novelists.

Question 16

Skill: Grammar / Conventions | Difficulty: Easy

The earliest public lending library in the English-speaking world opened during the reign of Queen Anne in 1701. Situated in the cathedral city of Norwich, the library was called the Norwich City Library. It was stocked with volumes that had been donated ______ the private collections of local clergy and merchants.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) from;
B) from
C) from:
D) from,

Question 17

Skill: Grammar / Conventions | Difficulty: Easy

The National Archaeological Museum of Athens, a research institution in the Attica region of Greece, is one of more than sixty museums in the country. Greece's museums ______ visitors about everything from the nation's classical heritage to its modern folk traditions.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) having educated
B) educating
C) to educate
D) educate

Question 18

Skill: Grammar / Conventions | Difficulty: Easy

In the spring of 1972, the album "Harvest" by Neil Young was a commercial sensation. Having remained on the charts for nine weeks, ______ reached the top position on the Billboard 200 list of best-selling albums.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) those
B) they
C) each
D) it

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

A community land trust is an organization that acquires and manages property for the benefit of its members. This model differs from conventional arrangements in which a single investor or developer controls a parcel. Because the revenue generated by a land trust is shared among all participants—who are also stakeholders—the members ______ directly from its growth.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) benefit
B) had benefited
C) benefited
D) were benefiting

Question 20

Skill: Grammar / Conventions | Difficulty: Medium

The 2018 documentary *Won't You Be My Neighbor?* was directed by Morgan Neville. It explores how Fred Rogers shaped children's television in the late twentieth ______ how the host's commitment to emotional honesty influenced an entire generation of young viewers and their families.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) century revealing
B) century; revealing
C) century, revealing
D) century. Revealing

Question 21

Skill: Transitions | Difficulty: Easy

Generally, insulated materials retain heat more effectively than uninsulated ones. For example, a double-walled stainless steel flask keeps its contents warm for hours with minimal temperature loss. ______ a thin ceramic mug radiates heat quickly, causing the liquid inside to cool rapidly.

Which choice completes the text with the most logical transition?

A) As a result,
B) Specifically,
C) In conclusion,
D) On the other hand,

Question 22

Skill: Transitions | Difficulty: Medium

In geology, a sediment layer deposited within a river delta is generally expected to remain near the surface if deposition continues; however, if deposition ceases, subsequent erosion may expose much older strata. The geologists Hana Okada and Leo Brandt, ______ documented a thick sequence of ancient clay beneath the modern surface deposits of the Ganges Delta, well below the active sediment layer that remains continuously replenished.

Which choice completes the text with the most logical transition?

A) fittingly,
B) for example,
C) likewise,
D) though,

Question 23

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Matsumoto is a city in Nagano Prefecture, Japan.
- Prefectures are administrative divisions responsible for providing many public services to their residents.
- One service they provide is disaster preparedness planning.
- Matsumoto covers an area of roughly 978 km^2.
- Nagano Prefecture is divided into 77 municipalities.

The student wants to emphasize the size of Matsumoto. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The municipality of Matsumoto in Nagano Prefecture, Japan, covers an area of roughly 978 km^2.
B) Providing disaster preparedness planning is just one example of the public services that municipalities provide.
C) Matsumoto—a municipality in Nagano Prefecture, Japan—provides many public services to its residents.
D) Matsumoto is one of 77 administrative regions, known as municipalities, across Nagano Prefecture.

Question 24

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Most of the amphibian and reptile species in Guam are non-native.
- In a 2021 survey, ecologists wanted to determine what role non-native lizards play in dispersing plant seeds in Guam.
- Researchers catalogued plant seeds found in droppings from non-native lizards.
- *Leucaena leucocephala*, a leguminous shrub, was one of twelve native species catalogued.
- *Coccinia grandis*, a climbing vine, was one of twenty-six non-native species catalogued.
- Researchers concluded that non-native lizards play a significant role in dispersing seeds of both native and non-native plants.

The student wants to emphasize a difference between the two plants. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Most plant species found in Guam, such as *Coccinia grandis*, are non-native.
B) Though *Leucaena leucocephala* and *Coccinia grandis* can both be found in Guam, only the former is native.
C) A 2021 survey catalogued plant seeds found in lizard droppings in Guam to determine the role of non-native lizards in seed dispersal.
D) Seeds from *Leucaena leucocephala* and *Coccinia grandis* plants were found in the droppings of non-native Guam lizards, according to a 2021 survey.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- An ice age is a prolonged period during which global temperatures drop significantly and ice sheets expand.
- Over Earth's history, ice ages are separated by warmer interglacial periods.
- This cyclical pattern is believed to span tens of thousands of years and is known as glacial cycling.
- The Karoo Ice Age was an ice age that began about 360 million years ago.
- The Quaternary Ice Age is the most recent ice age, beginning about 2.6 million years ago.

The student wants to emphasize the chronological order in which the ice ages occurred. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Karoo Ice Age began about 360 million years ago but eventually gave way to a warmer period.
B) The Karoo Ice Age and the Quaternary Ice Age were both prolonged periods of significant global cooling.
C) The Quaternary Ice Age began long after the Karoo Ice Age.
D) Spanning tens of thousands of years between them, ice ages are prolonged cold periods separated by warmer intervals.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Generally, a metal will expand when heated.
- The expansion of a metal due to heat is known as thermal expansion.
- A 2020 study led by Elena Sokolov and James Park tested the thermal expansion coefficients of various alloys.
- When a sample of brass alloy was heated, its length increased by 19.0 micrometers per degree Celsius.
- When a sample of Invar (a nickel-iron alloy) was heated, its length increased by only 1.2 micrometers per degree Celsius.

The student wants to emphasize a similarity between brass and Invar alloys. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2020, two research teams measured the thermal expansion coefficients of various alloys, including brass and Invar.
B) Both brass alloy and Invar expand when heated, according to a 2020 study.
C) Researchers determined that when the alloys were heated, the length of Invar increased far less than that of brass.
D) Heating a metal will generally cause its length to increase, a process known as thermal expansion.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- In a 2005 study, Romero and Vasquez tested the effect of mulch depth on weed suppression in vegetable gardens.
- The test site was a community garden in a temperate maritime climate in the Pacific Northwest.
- The researchers found that in these environmental conditions the presence of wood-chip mulch at a depth of 10 cm had a significant positive effect on weed suppression.
- Weed suppression occurs when a ground covering prevents weed seeds from receiving enough light to germinate.

Which choice most effectively uses information from the given sentences to present the study's findings to an audience already familiar with the concept of weed suppression?

A) The findings of Romero and Vasquez's study were published in 2005.
B) In a 2005 study by Romero and Vasquez, wood-chip mulch was found to have a positive effect on weed suppression, which occurs when a ground covering blocks light from reaching weed seeds.
C) The effect of mulch, which includes materials such as wood chips, on weed growth has been the subject of horticultural research.
D) Romero and Vasquez found that in a temperate maritime garden, the presence of wood-chip mulch at a depth of 10 cm significantly suppressed weeds.

MODULE 2 — EASIER PATH (27 Questions)

(For students scoring below ~60% on Module 1)

Question 1

Skill: Words in Context | Difficulty: Easy

Like many other coastal wetlands, the marshes along the Chesapeake Bay are ______: they shift in extent and composition as tidal patterns, storm surges, and sediment flows continually reshape the shoreline.

Which choice completes the text with the most logical and precise word or phrase?

A) static
B) immutable
C) volatile
D) permanent

Question 2

Skill: Words in Context | Difficulty: Easy

Aristophanes of Athens was a fifth-century BCE playwright who primarily wrote satirical comedies lampooning political figures of his era. His play *The Clouds*, however, is ______: featuring elaborate fictional scenarios involving celestial beings and philosophical academies, it is considered by some scholars as among the earliest works to blend comedy with speculative imagination.

Which choice completes the text with the most logical and precise word or phrase?

A) authorized
B) applicable
C) sarcastic
D) visionary

Question 3

Skill: Words in Context | Difficulty: Medium

For decades, neuroscientists assumed that adult human brains could not generate new neurons. Recent studies using carbon-14 dating of neural tissue, however, have provided evidence that neurogenesis does occur in certain brain regions throughout adulthood. These findings have proven ______ to researchers who had built their careers on the earlier consensus, forcing them to reconsider fundamental assumptions about brain plasticity.

Which choice completes the text with the most logical and precise word or phrase?

A) gratifying
B) disruptive
C) irrelevant
D) complementary

Question 4

Skill: Main Purpose | Difficulty: Easy

Established in 1987 in San Antonio, Texas, the Guadalupe Cultural Arts Center is devoted to preserving and promoting Chicano, Latino, and Native American art and culture. Since its founding, it has hosted more than 2,500 exhibitions and performances. More recently established US-based organizations dedicated to similar cultural missions include the National Museum of the American Latino, a Smithsonian affiliate. Based in Washington, D.C., it focuses on the art, history, and lived experiences of Latinos in the United States.

Which choice best states the main purpose of the text?

A) To trace the founding of two institutions, including how they acquired their initial collections.
B) To draw a contrast between the exhibition sizes of two cultural institutions.
C) To present information about two institutions, including each institution's area of focus.
D) To trace a historical development that encouraged the founding of two institutions.

Question 5

Skill: Overall Structure | Difficulty: Medium

Quilts produced by the women of Gee's Bend—a small, historically African American community in rural Alabama—are distinguished by their bold, improvisational geometric patterns: asymmetrical blocks of color, unexpected angles, and strips of contrasting fabric, to name a few elements. But there was room for personal expression within this shared tradition. Annie Mae Young's quilt *Denim Strips*, for instance, may at first glance resemble other Gee's Bend compositions, but its purposeful use of recycled work-clothing fabric gives it a textural and thematic dimension that other quilts in the tradition do not share.

Which choice best describes the overall structure of the text?

A) It describes an aesthetic framework shared by a particular group of artists and then makes and illustrates the claim that individuals introduced variations within that framework.
B) It describes the common perception that a particular group of artists' works are derivative and then provides a specific piece of evidence that reinforces that perception.
C) It offers historical context that accounts for a particular group of artists' shared style and then indicates the circumstances under which several members of that group began exploring more unconventional themes.
D) It explains how a particular group of artists began collaborating and then recounts how one member of that group became especially influential among them.

Question 6

Skill: Cross-Text | Difficulty: Hard

Text 1

Ocean thermal energy conversion (OTEC) systems exploit the temperature difference between warm surface water and cold deep water to generate electricity. In 2019, Kenji Tanaka et al. proposed a closed-cycle OTEC plant design for tropical Pacific islands, arguing that the consistently large thermal gradient (at least 20°C) in those waters makes commercial-scale OTEC viable. Their calculations showed that a plant at latitude 10°N could produce 50 MW continuously.

Text 2

In a 2022 analysis, Maria Espinoza et al. cautioned that closed-cycle OTEC systems require massive volumes of cold deep water, creating upwelling currents that may cool surface temperatures over time. Using oceanographic models, Espinoza et al. estimated that operating multiple OTEC plants within a 200 km radius could reduce the local thermal gradient by 3–5°C within a decade, potentially undermining the very conditions the technology depends on.

Based on the texts, how would Espinoza et al. (Text 2) most likely respond to Tanaka et al.'s research, as presented in Text 1?

A) By observing that unlike the OTEC design Tanaka et al. proposed, most other renewable energy systems for islands are likely located in temperate latitudes.
B) By maintaining that Tanaka et al. relied on a thermal gradient estimate that may not remain stable if multiple plants operate in the same region.
C) By arguing that OTEC plants in the tropical Pacific are unlikely to generate electricity because the islands are located too far from deep water.
D) By stating that the chemical composition of deep water in the region Tanaka et al. identified suggests that the water is unsuitable for closed-cycle OTEC systems.

Question 7

Skill: Inference / Complete the Text | Difficulty: Medium

Japanese architect Tadao Ando's prolific career, which spanned the 1970s to the 2020s, evolved through distinct phases. After visiting Europe and the United States in the late 1960s and studying the works of Le Corbusier, Ando began incorporating exposed concrete and geometric minimalism into his designs, as seen in the Row House in Sumiyoshi, whose stark unadorned surfaces contrast with his later projects in Osaka, such as the Church of the Light, which introduce dramatic interplays of natural light and shadow that evince a more contemplative, spiritual aesthetic.

Information in the text best supports which statement about the design of the Church of the Light?

A) It represents a transitional moment between the early and late phases of Ando's development.
B) It reflects an approach to light and atmosphere that Ando later abandoned.

C) It displays the effects of Ando's exposure to international architectural ideas in the 1960s.
D) It is characteristic of the austere minimalism that defined Ando's earliest work.

Question 8

Skill: Evidence - Support / Weaken | Difficulty: Hard

Coastal cities worldwide have reinforced their waterfronts with engineered barriers—seawalls, groins, and revetments—to mitigate erosion and storm damage, a practice known as shoreline armoring. To assess the effects of two types of armoring on intertidal invertebrate communities—groins and seawalls—Dr. Elena Marquez et al. surveyed clam, mussel, and barnacle populations at five sites along the Oregon coast. Using a Benthic Community Health Index (BCHI), on which a high score corresponds to high community health, the researchers found that seawalls are more strongly negatively associated with intertidal invertebrate health than groins are.

Which finding, if true, would most directly illustrate the researchers' finding?

A) Invertebrate communities at Cannon Beach, a site with a high percentage of shoreline consisting of seawalls and groins, had lower average BCHI scores than did invertebrate communities at Gold Beach, a site with a low percentage of shoreline consisting of seawalls and groins.
B) Invertebrate communities at Newport, a site with a relatively high percentage of shoreline consisting of seawalls, had lower average BCHI scores than did invertebrate communities at Bandon, a site with a relatively high percentage of shoreline consisting of groins.
C) Invertebrate communities at Cannon Beach, a site with equal percentages of shoreline consisting of seawalls and groins, had higher average BCHI scores than did invertebrate communities at Bandon, a site with different percentages of shoreline consisting of seawalls and groins.
D) The difference in average BCHI scores for invertebrate communities at Seaside and Newport, two sites with a higher percentage of shoreline consisting of seawalls than of groins, was statistically insignificant.

Question 9

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Graph: A line graph titled "Lake Mendota Ice Cover Duration 2010–2020." The x-axis shows years from 2010 to 2020. The y-axis shows "Days of ice cover" from 0 to 130. Three lines are plotted: a solid line with triangle markers for "total ice cover days," a dashed line with square markers for

“clear-ice days,” and a dotted line with circle markers for “snow-ice days.” From 2010 to 2017, total ice cover days fluctuated between 80 and 120, with clear-ice and snow-ice roughly equal. In 2018, total ice cover dropped to about 70, driven mainly by a sharp decline in snow-ice days (from ~55 to ~25), while clear-ice days remained around 45. In 2019 and 2020, total ice days remained low (~65–75).]

Lake Mendota ice cover is an important ecological indicator for Wisconsin’s freshwater habitats. Historically, snow-ice (formed when snow saturates and refreezes on the surface) and clear-ice (formed directly from lake water) have contributed roughly equally to total cover duration. Although declining total ice duration has been linked to rising average winter air temperatures, researchers note that the composition of ice types matters as well, since snow-ice insulates the water column differently than clear-ice does.

Which statement, if true, would account for data shown in the graph and would illustrate the point made by the researchers?

A) In early 2018, unusually warm January rains fell on the lake surface, washing away accumulated snow and preventing the formation of snow-ice while having little effect on clear-ice formation.
B) In early 2017, a fungal bloom in the lake reduced dissolved oxygen levels but did not affect ice formation processes.
C) Between 2010 and 2016, both clear-ice days and total ice days increased steadily as average winter temperatures declined.
D) Winter air temperatures in the region rose gradually from 2010 to 2017, but in early 2018 there was an unprecedented warm spell that disrupted both clear-ice and snow-ice formation equally.

Question 10

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

Recorded Annual Migration Distances for Four Animal Species

Species	Region	Range (km)	Tracking Method
Monarch butterfly	North America	4,800	Tag-recapture
Arctic tern	Global	71,000	GPS
Humpback whale	Pacific	8,300	Photo ID
Wildebeest	Africa	1,600	GPS

Some studies of migrating animals measure the distance an organism travels by fitting individuals with GPS transmitters. Other studies use indirect methods like tag-recapture, which estimates range based on the farthest points at which the same individual is documented. A researcher argues that because GPS records continuous movement, using GPS to track a species would yield significantly longer recorded distances than tag-recapture would. For example, it is very likely that the distance reported for the ______

Which choice most effectively uses data from the table to complete the example?

A) Arctic tern would be less than 8,300 kilometers if the photo ID method had been used.
B) humpback whale would be less than 71,000 kilometers if the GPS method had been used.
C) wildebeest would be greater than 4,800 kilometers if the tag-recapture method had been used.
D) monarch butterfly would be greater than 71,000 kilometers if the GPS method had been used.

Question 11

Skill: Evidence - Support / Weaken | Difficulty: Medium

Thermoelectric generators convert thermal gradients into electrical voltage, eliminating the need for moving parts. The vibration of heavy machinery in a factory, for instance, produces enough waste heat to power several onboard sensors thermoelectrically. A newly engineered thermoelectric module incorporating a highly efficient bismuth telluride (Bi_2Te_3) semiconductor has been demonstrated to sustain consistent power output under fluctuating heat loads, an essential requirement for industrial monitoring devices to function reliably.

Which finding, if true, would most directly support the text's claim about industrial monitoring devices?

A) The near-constant thermal output of an electric furnace makes it possible to power its monitoring sensors using only standard, non-Bi_2Te_3 thermoelectric modules.
B) Irregular or fluctuating power supply compromises the accuracy of industrial monitoring sensors.
C) The Bi_2Te_3 semiconductor is incompatible with most existing industrial monitoring devices.
D) The high efficiency of the Bi_2Te_3 module is what makes the energy output from a thermoelectric generator adequate for industrial monitoring devices.

Question 12

Skill: Inference / Complete the Text | Difficulty: Medium

Shoppers increasingly expect that items they order online will arrive within hours, not days. Although improvements in warehouse automation have greatly accelerated order processing, last-kilometer delivery (the final leg of transport to the customer's doorstep) continues to pose a bottleneck for logistics companies. Time constraints imposed by same-day guarantees are not the sole hurdle; other obstacles, such as traffic congestion and limited parking in urban centers, persist. While innovations to address these challenges have been tested—the use of cargo bicycles in dense neighborhoods, for instance—adoption has been limited because of additional complications that arise (e.g., cargo bicycles cannot carry items exceeding certain weight limits). Consequently, ______

Which choice most logically completes the text?

A) in the near term, logistics companies are unlikely to overcome the impediments associated with last-kilometer delivery.

B) innovations in last-kilometer delivery seem poised to increase shoppers' expectations for same-day shipping.

C) logistics companies should invest more resources in proven warehouse technologies than in experimental last-kilometer solutions.

D) the use of cargo bicycles may enable logistics companies to meet consumer expectations now but likely is not viable as a permanent solution.

Question 13

Skill: Inference / Complete the Text | Difficulty: Medium

Paleontologists searching for evidence of early vertebrates in Africa have uncovered many well-preserved fossils of ancient fish, including in a geological formation at the Strud site in Morocco. However, to find even earlier specimens such as primitive jawless fish from the Ordovician period, researchers must look elsewhere, such as in the Ordovician-age sedimentary deposits at Erfoud in southeastern Morocco, because ______

Which choice most logically completes the text?

A) Africa has sites with ancient fish fossils but not sites with primitive jawless organisms.

B) primitive jawless fish are found at a broader range of global sites than ancient jawed fish.

C) those deposits are from a significantly earlier era than the formation at the Strud site.

D) the jawless fish at Erfoud are better preserved than the jawed fish at the Strud site.

Question 14

Skill: Inference / Complete the Text | Difficulty: Hard

Solid-oxide fuel cells (SOFCs) harness the movement of oxygen ions through a ceramic electrolyte to generate electricity from a fuel source. The fuel—typically hydrogen or methane

—reacts at the anode, releasing electrons that travel through an external circuit to the cathode. However, transferring oxygen ions efficiently across the electrolyte requires that the ions pass through a crystalline lattice with minimal resistance. Typical ceramic electrolytes achieve acceptable conductivity only at very high temperatures (above 800°C). In an experiment, researchers substituted a samarium-doped ceria (SDC) electrolyte into an SOFC. The resulting device achieved comparable power output at just 550°C. Since materials with more open crystal lattices allow ions to migrate with less thermal energy, the researchers hypothesized that ______

Which choice most logically completes the text?

A) SDC nanoparticles may enhance the chemical reaction rate at the anode, thereby increasing the number of free electrons available for the external circuit.
B) as the operating temperature of an SOFC increases, the crystalline lattice of the electrolyte may contract, independent of the type of ceramic used.
C) oxygen ions may be conducted through the electrolyte before the SDC lattice reaches its optimal operating temperature.
D) SDC may possess a more open crystalline structure than standard ceramic electrolytes, allowing oxygen ions to pass through with less thermal energy.

Question 15

Skill: Inference / Complete the Text | Difficulty: Hard

The concentration of dissolved silica in ocean water—expressed as a ratio of silica molecules to total dissolved solids—affects the growth patterns of diatoms, a group of single-celled algae. For the Southern Ocean, the observational records of silica concentrations are limited and inconsistent, making it difficult to determine the exact range of dissolved silica at various depths. Clara Voss and colleagues note that the outputs of the GEOTRACES biogeochemical model, which generally align with observations in several respects, are based on prescribing a uniform silica concentration for the upper 200 meters of the water column. It is therefore important to note that ______

Which choice most logically completes the text?

A) further sampling of the Southern Ocean may define the silica concentration precisely enough for the model to replace its uniform estimate with a depth-varying profile.
B) some discrepancies between the model's predictions of diatom density and actual diatom density in the Southern Ocean are to be expected.
C) variations in the model's simulations of diatom growth at different depths could be traceable to differences in dissolved silica concentration.
D) even though the model's outputs sometimes match observational data, the actual dissolved silica in the Southern Ocean is likely higher than the concentration used in the model.

Question 16

Skill: Grammar / Conventions | Difficulty: Easy

Yayoi Kusama, a Japanese artist and sculptor, created many large-scale immersive installations, such as her series Infinity Mirror Rooms. In these works, Kusama's signature aesthetic is typified by her use of polka-dot ______ often incorporate reflective surfaces and LED lights drawn from science and technology.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) patterns. That
B) patterns: that
C) patterns; that
D) patterns that

Question 17

Skill: Grammar / Conventions | Difficulty: Easy

Capoeira, a martial art that originated in Brazil, has become an international ______ 2019, the French duo known as Lucas and Marie defeated competitors from around the world to win the annual World Capoeira Championship held in Salvador, Brazil, becoming the first European champions.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) phenomenon in
B) phenomenon, in
C) phenomenon and in
D) phenomenon. In

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

The Dravidian language family encompasses some 80 ______ Kannada and Malayalam, which are spoken by 44 million and 38 million speakers, respectively—and accounts for nearly a quarter of India's total population of speakers, making it of great interest to linguists like Bhadriraju Krishnamurti.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) languages,
B) languages:
C) languages
D) languages—

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

Although Halley's Comet and Comet Hale-Bopp are both classified as long-period comets—celestial objects with orbital periods exceeding 200 years—they exhibit striking differences in ______ Halley's Comet is relatively predictable, returning to the inner solar system approximately every 76 years, while Hale-Bopp's orbit is so elongated that its next perihelion will not occur for roughly 2,500 years.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) periodicity; the
B) periodicity.
C) periodicity, while the
D) periodicity: while the

Question 20

Skill: Grammar / Conventions | Difficulty: Medium

Chloroplast genomes replicate semi-autonomously, which should over successive generations lead to an accumulation of deleterious mutations and a decline in photosynthetic efficiency. However, nuclear genes are thought to coevolve with the rate of chloroplast deterioration, correcting mutational errors. Such a corrective mechanism ______ the organelle's decline, unlike enzyme-stabilizing chaperone proteins, offers an evolutionary explanation for the persistence of chloroplast function.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had counteracted
B) counteracted
C) counteracts
D) counteracting

Question 21

Skill: Grammar / Conventions | Difficulty: Medium

In a national referendum, one might expect voter turnout to be highest in the most densely populated provinces. However, if pre-election surveys and historical records suggest that the outcome in a given province is a foregone conclusion, turnout in that province will likely be lower. Ultimately, a province's historical voting patterns and demographic engagement data, not its population size, ______ its turnout rates in referenda.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) determines
B) has determined
C) determining
D) determine

Question 22

Skill: Transitions | Difficulty: Easy

To appreciate the gradual shifts in Japanese woodblock printing technique during the Edo period, first consider Hishikawa Moronobu's monochrome prints from the 1670s; ______ compare them to Suzuki Harunobu's full-color nishiki-e prints from the 1760s.

Which choice completes the text with the most logical transition?

A) therefore,
B) still,
C) instead,
D) next,

Question 23

Skill: Transitions | Difficulty: Medium

Soil contaminated with cadmium (a toxic heavy metal) is harmful to most crops and livestock, but the plant species *Thlaspi caerulescens*, or alpine pennycress, not only tolerates such conditions but also helps decontaminate them. As a metal hyperaccumulator, *Thlaspi caerulescens* absorbs large quantities of cadmium and stores it in its leaves and stems; ______ cadmium concentrations in the surrounding soil decrease.

Which choice completes the text with the most logical transition?

A) accordingly,
B) specifically,
C) nevertheless,
D) in addition,

Question 24

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- The Japanese word “yen” derives from the Chinese word “yuan.”
- Today, more than fifteen different Asian currencies include the word “yuan” or a cognate in their names.
- Japan’s currency is known as the Japanese yen.

The student wants to provide an example of a country that uses a cognate of “yuan” in the name of its currency. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Japan uses the yen as its national currency.
B) Japan does not refer to its currency by the Chinese word “yuan.”
C) The Chinese word “yuan” is the origin of the Japanese word “yen.”
D) Today, many Asian countries use a form of the word “yuan” in naming their currencies.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Formalist analysis and biographical analysis are two approaches to literary criticism.
- Formalist analysis examines how a text’s structural features contribute to its overall meaning.
- Such an analysis of James Joyce’s *Ulysses* might consider how the novel’s stream-of-consciousness technique mirrors the fragmentation of modern urban life.
- Biographical analysis considers the life circumstances of the author when interpreting a work.
- Such an analysis of Virginia Woolf’s *Mrs Dalloway* might consider how the author’s experiences with mental illness informed the novel’s portrayal of trauma.

The student wants to present biographical analysis to an audience unfamiliar with the concept. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) An approach to literary criticism, biographical analysis considers the life circumstances of the author when interpreting a work.
B) *Mrs Dalloway*'s portrayal of trauma can be linked to Woolf's own mental health experiences.
C) A formalist analysis of *Ulysses* might examine how stream-of-consciousness technique reflects modern fragmentation.
D) Biographical analysis differs from formalist analysis in that formalist analysis examines how a text's structural features contribute to its meaning.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Silicon can exist in several distinct structural forms called polymorphs.
- Quartz is a silicon polymorph.
- Quartz is composed of silicon and oxygen atoms arranged in a continuous framework of SiO_4 tetrahedra.
- Silicon nanowires are silicon structures at the nanoscale.
- They are composed of silicon atoms arranged into elongated cylindrical shapes.

The student wants to emphasize a difference between silicon nanowires and quartz. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Composed of atoms arranged into elongated cylinders, silicon nanowires contrast with quartz, a silicon polymorph composed of a continuous SiO_4 framework.
B) Quartz and silicon nanowires are both silicon-based structures, but silicon can exist in several distinct forms.
C) Unlike silicon nanowires, which are composed of silicon atoms arranged in elongated shapes, quartz is a cylindrical silicon polymorph.
D) The SiO_4 tetrahedra in quartz are stacked, whereas in silicon nanowires, they form elongated tubes.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- Bounded rationality is the idea that cognitive limitations constrain decision-making.
- 1955: economist Herbert Simon introduced the concept of bounded rationality.
- In Simon's framework, individuals cannot process all available information and therefore rely on simplified strategies (heuristics).
- 2002: psychologist Gerd Gigerenzer proposed the adaptive toolbox hypothesis.
- The adaptive toolbox hypothesis argues that heuristics are not flawed shortcuts but efficient strategies well-suited to specific environments.

The student wants to compare Simon's framework with Gigerenzer's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2002, Gigerenzer proposed the adaptive toolbox hypothesis, disagreeing with Simon's earlier claim that cognition is constrained by biological neural architecture.
B) In describing heuristics as efficient rather than merely simplifying, Gigerenzer's adaptive toolbox hypothesis offers a more positive interpretation of the strategies Simon attributed to bounded rationality.
C) The frameworks of Simon and Gigerenzer differ in whether they consider low-level cognitive processes, such as perception and memory, bounded.
D) Following Simon's 1955 concept of bounded rationality, Gigerenzer proposed that cognitive constraints apply to all decision-making domains.

MODULE 2 — HARDER PATH (27 Questions)

(For students scoring above ~60% on Module 1)

Question 1

Skill: Words in Context | Difficulty: Hard

The oral histories preserved by the Maori iwi (tribal groups) of Aotearoa New Zealand are not mere retellings of ancestral events; they serve as _____ records—encoding genealogical connections, land rights, and political alliances—that continue to carry legal and cultural authority in contemporary treaty negotiations.

Which choice completes the text with the most logical and precise word or phrase?

A) provisional
B) ornamental
C) speculative
D) juridical

Question 2

Skill: Words in Context | Difficulty: Hard

While many conservation biologists have assumed that habitat fragmentation uniformly diminishes species diversity, a meta-analysis led by Lenore Fahrig found that the relationship is far more ______: in nearly half the studies reviewed, fragmentation per se—independent of habitat loss—was associated with neutral or even positive effects on biodiversity, challenging the field's dominant paradigm.

Which choice completes the text with the most logical and precise word or phrase?

A) equivocal
B) negligible
C) corrosive
D) transparent

Question 3

Skill: Words in Context | Difficulty: Medium

Although the biennial Venice Architecture Exhibition attracts global attention, the structures displayed there are typically temporary pavilions that are dismantled after the event. The exhibition's influence is therefore largely ______: rather than reshaping the built environment directly, it sparks debates and conceptual experiments that filter into architectural practice over subsequent years.

Which choice completes the text with the most logical and precise word or phrase?

A) tangible
B) indirect
C) instantaneous
D) diminishing

Question 4

Skill: Main Purpose | Difficulty: Hard

In a 2021 paper, historian Mariana Flores argues that standard accounts of the Mexican Revolution (1910–1920) overemphasize military campaigns at the expense of the quotidian resistance practiced by rural women. Flores's archival research in Morelos reveals that women organized clandestine supply networks, sheltered displaced families, and maintained agricultural production during periods of intense combat. These activities, Flores contends, were not ancillary to the revolutionary effort but constituted a parallel front without which the armed movements would have collapsed. By foregrounding these contributions, Flores seeks to revise the conventional historiography that frames the revolution primarily as a sequence of battles and political negotiations.

Which choice best states the main purpose of the text?

A) To argue that rural women's contributions to the Mexican Revolution were more important than those of soldiers.
B) To evaluate the relative military effectiveness of two different strategies employed during the Mexican Revolution.
C) To compare Flores's archival methodology with that of earlier historians who studied the Mexican Revolution.
D) To describe a historian's argument that the conventional narrative of a major event overlooks essential contributions by a particular group.

Question 5

Skill: Overall Structure | Difficulty: Hard

The following text is adapted from a 1922 essay by Virginia Woolf.

To read a novel is, some hold, a passive occupation: one sits in an armchair, turns pages, and absorbs. But this view is hopelessly crude. The reader who deserves the name is perpetually active—constructing, questioning, discarding hypotheses at a speed the author could never have foreseen. Consider that a single metaphor on page forty may recolour everything that came before it, obliging the reader to return, to revise, to rebuild the architecture of understanding. The author supplies materials; the reader is the architect.

Which choice best describes the overall structure of the text?

A) It introduces a widely held view, rejects that view, and then supports the rejection with an example illustrating the complexity of the activity in question.
B) It presents two competing theories about reading, evaluates each theory's merits, and concludes that neither is entirely adequate.
C) It describes a historical change in attitudes toward reading, identifies the cause of that change, and predicts its future trajectory.
D) It poses a question about the nature of reading, surveys several possible answers, and selects the one best supported by literary scholarship.

Question 6

Skill: Cross-Text | Difficulty: Hard

Text 1

Soil microbiome transplants—introducing microbial communities from healthy soil into degraded land—have been proposed as a restoration tool. In a 2020 field trial, Jia Wei et al. inoculated depleted agricultural plots in Hebei Province with microbiome slurries from adjacent grasslands. After two growing seasons, transplanted plots showed 35% greater root biomass and significantly higher populations of nitrogen-fixing bacteria compared to untreated controls.

Text 2

In a 2023 greenhouse study, Amara Osei et al. replicated soil microbiome transplants under controlled conditions but observed that the introduced microbial communities were almost entirely displaced by the resident microbiome within eight weeks. Osei et al. hypothesize that the competitive dominance of established microbial networks may render transplants ineffective once the initial inoculum is consumed, suggesting that Wei et al.'s field results may reflect transient nutrient pulses from the slurry itself rather than lasting microbial colonization.

Based on the texts, how would Osei et al. (Text 2) most likely respond to Wei et al.'s findings (Text 1)?

A) By maintaining that soil microbiome transplants are effective in field settings but not in greenhouses due to differences in light exposure.

B) By suggesting that Wei et al.'s experimental design was flawed because it failed to include a control group.

C) By arguing that the increased root biomass observed by Wei et al. was likely caused by a temporary nutrient boost rather than sustained microbial establishment.

D) By proposing that the nitrogen-fixing bacteria in Wei et al.'s study were indigenous to the agricultural plots and not introduced by the transplant.

Question 7

Skill: Inference / Complete the Text | Difficulty: Hard

Economist Ha-Joon Chang has argued that the standard narrative of free-trade-driven industrialization misrepresents the historical record. In *Kicking Away the Ladder* (2002), Chang demonstrates that virtually every nation now considered an advanced economy—including Britain, the United States, and Germany—relied on protectionist tariffs, subsidies, and state-directed investment during its own period of industrial development. Only after these nations had established dominant positions in global markets did they begin advocating for the free-trade policies they now urge developing countries to adopt. Chang concludes that prescribing immediate trade liberalization to developing economies effectively ______

Which choice most logically completes the text?

A) validates the theoretical models that predict mutual gains from unrestricted trade between nations at similar levels of development.
B) denies those economies access to the same developmental strategies that today's wealthy nations once employed.
C) overstates the role that domestic policy plays relative to geographic and climatic advantages in determining economic growth.
D) encourages developing economies to specialize in industries where they already possess a comparative advantage.

Question 8

Skill: Evidence - Support / Weaken | Difficulty: Hard

Recent studies on the gut-brain axis suggest that the composition of intestinal microbiota can influence mood and cognitive function through the production of neurotransmitter precursors. Neuroscientist Keiko Tanaka and colleagues hypothesize that a specific bacterial strain, *Lactobacillus rhamnosus*, may reduce anxiety-like behavior in mice by increasing gamma-aminobutyric acid (GABA) signaling in the brain.

Which finding, if true, would most directly support the researchers' hypothesis?

A) Mice colonized with *L. rhamnosus* show elevated GABA receptor expression in the prefrontal cortex and reduced anxiety-like behavior compared to germ-free controls.
B) Mice raised in germ-free environments display higher baseline anxiety than conventionally raised mice regardless of which bacterial strains are later introduced.
C) Administration of *L. rhamnosus* to mice increases intestinal serotonin production but does not affect GABA levels in the brain.
D) Anxiety-like behavior in mice decreases following administration of a broad-spectrum antibiotic that eliminates most gut bacteria, including *L. rhamnosus*.

Question 9

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

GRAPH TO INSERT

[Graph: A multi-line graph titled "Urban Tree Canopy Coverage in Three Neighborhoods, 2005–2020." X-axis: years 2005, 2010, 2015, 2020. Y-axis: percentage of land area covered by tree canopy (0% to 40%). Three lines: Neighborhood A (solid) starts at ~30% in 2005 and declines steadily to

~18% in 2020; Neighborhood B (dashed) starts at ~15% in 2005, rises to ~22% in 2015, then drops sharply to ~12% in 2020; Neighborhood C (dotted) starts at ~10% in 2005 and rises steadily to ~25% in 2020.]

Urban ecologists studying heat-island mitigation have linked tree canopy coverage to lower ambient surface temperatures. While some neighborhoods show canopy gains from tree-planting programs, others experience losses from disease, development, or severe weather events.

Based on the graph, which of the following statements is most accurate?

A) All three neighborhoods experienced net canopy loss between 2005 and 2020.
B) Neighborhood B's canopy coverage increased consistently throughout the entire period.
C) By 2020, Neighborhood C had achieved a higher canopy percentage than Neighborhood A, despite starting with less than half of A's coverage in 2005.
D) Neighborhood A maintained the highest canopy coverage of the three neighborhoods at every point measured.

Question 10

Skill: Inference / Complete the Text | Difficulty: Hard

The ratio of carbon-13 to carbon-12 isotopes in marine sediment cores—expressed as the $\delta^{13}C$ value—serves as a proxy for ocean productivity over geological timescales. For the Cretaceous period, the available $\delta^{13}C$ records are sparse and sometimes contradictory, making it difficult to reconstruct productivity trends with precision. Paleoceanographer Rui Chen and colleagues observe that the outputs of the COPSE Earth system model, which reproduce many features of the geological record, assume a constant $\delta^{13}C$ value for the shallow-ocean reservoir. It is therefore important to recognize that ______

Which choice most logically completes the text?

A) additional sampling of Cretaceous sediment cores may narrow the $\delta^{13}C$ range enough for the model to adopt a time-varying input.
B) even though the model's outputs occasionally match proxy records, the actual Cretaceous $\delta^{13}C$ was probably higher than the value used in the model.
C) discrepancies within the model's own simulations of Cretaceous ocean productivity could result from variations in the assumed $\delta^{13}C$ value.
D) some disagreements between the model's productivity predictions and observed proxy data from the Cretaceous are expected.

Question 11

Skill: Inference / Complete the Text | Difficulty: Hard

Lithium-sulfur (Li-S) batteries promise significantly higher energy density than conventional lithium-ion cells, but a major obstacle to their commercialization is the polysulfide shuttle effect: intermediate lithium polysulfide species dissolve into the electrolyte, migrate to the anode, and cause irreversible capacity loss. In a 2023 experiment, researchers coated the cathode with a metal-organic framework (MOF) membrane. The resulting battery retained 92% of its initial capacity after 500 cycles, compared with 64% for an uncoated cathode. Since MOF membranes are known for their precisely sized molecular pores, the researchers concluded that ______

Which choice most logically completes the text?

A) lithium polysulfides may accumulate within the MOF pores, eventually saturating the membrane and restoring the shuttle effect.
B) the MOF membrane likely acts as a selective barrier, permitting lithium-ion transport while physically blocking the larger polysulfide molecules from migrating to the anode.
C) the high surface area of MOF membranes may catalyze the polysulfide reactions at the cathode, increasing the rate at which polysulfides form.
D) conventional lithium-ion batteries may also benefit from MOF coatings, since they experience analogous dissolution problems at the cathode.

Question 12

Skill: Grammar / Conventions | Difficulty: Easy

The Mesoamerican ballgame, which was played by Indigenous civilizations across Central America for over three thousand years, held deep ritual ______ archaeological evidence including stone carvings and ceramic figurines depicts players in ceremonial regalia, suggesting the game carried cosmological meaning.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) significance, the
B) significance. The
C) significance; the
D) significance the

Question 13

Skill: Grammar / Conventions | Difficulty: Medium

The Trans-Siberian Railway, completed in 1916 after more than two decades of ______ connects Moscow to Vladivostok across 9,289 kilometers, making it the longest railway line in the world.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) construction;
B) construction:
C) construction,
D) construction

Question 14

Skill: Grammar / Conventions | Difficulty: Hard

Ribosomal RNA sequences diverge at measurable rates over evolutionary timescales, which should in principle allow biologists to estimate when two lineages shared a common ancestor. However, the molecular clock for rRNA is hypothesized to vary across taxa, with rapidly evolving lineages accumulating substitutions faster than slowly evolving ones. Such rate variation, unlike phylogenetic noise from horizontal gene transfer, ______ a systematic bias into divergence-time estimates that standard tree-building algorithms cannot easily correct.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had introduced
B) introducing
C) introduces
D) introduced

Question 15

Skill: Grammar / Conventions | Difficulty: Hard

In a multi-party parliamentary system, one might expect coalition governments to form around the largest parties. However, if pre-election polling and historical precedent suggest that a particular coalition is ideologically incoherent, party leaders will avoid it. Ultimately, a party's programmatic alignment and coalition history, not its seat count, ______ its attractiveness as a coalition partner.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) determines
B) has determined
C) determining

D) determine

Question 16

Skill: Grammar / Conventions | Difficulty: Hard

Linguist Noam Chomsky's theory of universal grammar posits that all human languages share a common structural ______ the theory has prompted decades of cross-linguistic research aimed at identifying the specific syntactic parameters that vary between languages.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) foundation. And
B) foundation and
C) foundation; and
D) foundation, and

Question 17

Skill: Transitions | Difficulty: Medium

To trace the evolution of Chinese landscape painting, first examine Fan Kuan's monumental hanging scroll *Travelers Among Mountains and Streams* from around 1000 CE; ______ compare it to Ni Zan's spare, austere compositions from the fourteenth century.

Which choice completes the text with the most logical transition?

A) therefore,
B) still,
C) instead,
D) then,

Question 18

Skill: Transitions | Difficulty: Hard

Groundwater contaminated with arsenic (a toxic metalloid) poses serious health risks in many South Asian communities, but the fern species *Pteris vittata*, or brake fern, not only tolerates arsenic-rich soil but also actively extracts the contaminant. As a hyperaccumulator, *Pteris vittata* sequesters arsenic in its fronds at concentrations hundreds of times greater than the surrounding soil; ______ arsenic levels in the remediated soil gradually decline to within safe thresholds.

Which choice completes the text with the most logical transition?

A) specifically,
B) consequently,
C) nevertheless,
D) in addition,

Question 19

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- Epistemic injustice is a concept describing wrongs related to knowledge and credibility.
- 2007: philosopher Miranda Fricker introduced testimonial injustice, a form of epistemic injustice in which a speaker receives less credibility due to prejudice against their social identity.
- In Fricker's framework, the listener's prejudice is the primary mechanism of harm.
- 2011: philosopher Kristie Dotson introduced the concept of testimonial smothering.
- Testimonial smothering occurs when a speaker self-censors because they anticipate that their audience will not give their testimony uptake.
- Dotson's concept shifts the focus from the listener's prejudice to the speaker's constrained agency.

The student wants to compare Fricker's concept with Dotson's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2011, Dotson introduced testimonial smothering, disagreeing with Fricker's earlier claim that epistemic injustice is primarily a structural phenomenon.
B) While Fricker's testimonial injustice locates harm in a listener's prejudiced credibility judgments, Dotson's testimonial smothering highlights how speakers may preemptively silence themselves in anticipation of such prejudice.
C) The concepts of Fricker and Dotson differ in whether they consider institutional policies, such as hiring practices, to be forms of epistemic injustice.
D) Following Fricker's 2007 introduction of testimonial injustice, Dotson proposed that all forms of epistemic injustice involve both speaker self-censorship and listener prejudice.

Question 20

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- Fermentation and respiration are two metabolic pathways by which organisms extract energy from glucose.
- Fermentation is anaerobic; it does not require oxygen.
- In ethanol fermentation (used by yeast), glucose is converted to ethanol and CO_2, yielding 2 ATP per glucose molecule.
- Aerobic respiration requires oxygen.
- In aerobic respiration, glucose is fully oxidized to CO_2 and H_2O, yielding approximately 36 ATP per glucose molecule.

The student wants to emphasize the difference in energy yield between the two pathways. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both fermentation and aerobic respiration convert glucose into usable energy, though through different biochemical mechanisms.

B) Yeast cells perform ethanol fermentation, an anaerobic process that produces ethanol, CO_2, and ATP.

C) While ethanol fermentation generates only 2 ATP per glucose molecule, aerobic respiration yields roughly 36 ATP—an 18-fold difference in energy efficiency.

D) Aerobic respiration requires oxygen, whereas fermentation can proceed without it, making fermentation advantageous in low-oxygen environments.

Question 21

Skill: Inference / Complete the Text | Difficulty: Hard

Political scientist Elinor Ostrom challenged the prevailing assumption in her discipline that common-pool resources—fisheries, irrigation systems, forests—will inevitably be overexploited unless managed by either a centralized government authority or privatized into individual holdings. Through decades of fieldwork in communities across Nepal, Spain, and the Philippines, Ostrom documented numerous cases in which local users devised and enforced their own governance rules that sustained the resource over centuries. Her work implies that the conventional dichotomy between state control and privatization ______

Which choice most logically completes the text?

A) overlooks the capacity of communities to construct effective self-governing institutions.

B) accurately describes the situation in most developing countries where common-pool resources exist.

C) applies primarily to fisheries and irrigation systems rather than to forests.
D) is supported by the evidence Ostrom collected in Nepal but undermined by her data from Spain.

Question 22

Skill: Evidence - Support / Weaken | Difficulty: Hard

Archaeologists studying the collapse of the Classic Maya civilization (circa 800–1000 CE) have long debated whether prolonged drought was a primary causal factor. Paleoclimatologist Douglas Kennett and colleagues analyzed oxygen isotope ratios in stalagmites from Yok Balum Cave in Belize and found that the driest intervals in the record correspond closely with periods of intensified warfare and political fragmentation documented in Maya inscriptions. Kennett's team hypothesizes that drought-induced crop failures strained existing tribute systems, triggering elite competition and eventual state dissolution.

Which finding, if true, would most directly weaken Kennett's hypothesis?

A) Several Maya polities that experienced severe drought during the Terminal Classic period nevertheless maintained stable political institutions for an additional century.
B) The oxygen isotope record from Yok Balum Cave is consistent with other paleoclimate proxies from the region.
C) Maya agricultural techniques included sophisticated water management systems that could partially buffer communities against short-term rainfall deficits.
D) Inscriptions from the Late Classic period indicate that inter-polity warfare was already common during periods of above-average rainfall.

Question 23

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

[Graph description: A stacked area chart titled "Global Primary Energy by Source, 1990–2020." The x-axis shows years (1990, 2000, 2010, 2020). The y-axis shows exajoules (0 to 600). Four stacked areas from bottom to top: Fossil fuels (dark, largest area, ~450 EJ in 1990 growing to ~510 EJ in 2020), Nuclear (medium gray, ~20 EJ, relatively flat), Hydropower (light gray, ~25 EJ in 1990, ~40 EJ in 2020), Other renewables (lightest, ~5 EJ in 1990, growing steeply to ~50 EJ in 2020). Total energy demand grows from ~500 EJ to ~600 EJ.]

Energy analysts examining the global energy transition note that while renewable sources (excluding hydropower) have grown rapidly in absolute terms since 1990, fossil fuels continue to supply the majority of the world's primary energy. The analysts caution that growth in total energy demand has largely offset gains in renewable capacity.

Which statement is best supported by the data described in the graph?

A) By 2020, other renewables had surpassed nuclear energy in absolute contribution to global primary energy.
B) Fossil fuel consumption declined in absolute terms between 2010 and 2020.
C) Hydropower's share of total energy supply increased from 1990 to 2020.
D) Other renewables accounted for the majority of the increase in total energy supply between 1990 and 2020.

Question 24

Skill: Inference / Complete the Text | Difficulty: Hard

The concept of "shifting baselines" in ecology—first articulated by Daniel Pauly in 1995—describes the phenomenon in which each new generation of scientists accepts the environmental conditions they first encounter as "normal," failing to recognize long-term degradation that occurred before their careers began. This cognitive bias has practical consequences: if contemporary marine biologists assess fish stock health by comparison with populations measured twenty years ago rather than two hundred years ago, their conservation targets may be calibrated to an already depleted state. The concept therefore suggests that ______

Which choice most logically completes the text?

A) shifting baselines are primarily a concern in marine biology and have limited relevance to terrestrial ecosystems.
B) marine biologists should focus exclusively on the most recent data because older records are unreliable.
C) historical ecological data, even when imprecise, may be essential for setting accurate restoration benchmarks.
D) conservation targets based on recent data are more achievable and therefore more useful than those based on historical records.

Question 25

Skill: Grammar / Conventions | Difficulty: Hard

The philosopher Simone de Beauvoir drew a sharp distinction between two forms of freedom: abstract freedom, which exists as a theoretical ______ situated freedom, which is constrained by the material, social, and historical circumstances in which an individual actually lives.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) possibility, and
B) possibility and
C) possibility; and
D) possibility. And

Question 26

Skill: Grammar / Conventions | Difficulty: Hard

Permafrost soils in the Arctic contain vast stores of organic carbon that, if thawed, could decompose and release greenhouse gases. Climate models project that rising temperatures will cause widespread permafrost degradation; however, the rate at which thawed carbon ______ to atmospheric CO_2 depends on soil moisture, microbial community composition, and oxygen availability—variables that current models represent only coarsely.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) is converted
B) converting
C) having converted
D) to convert

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching a topic, a student has taken the following notes:

- Antibiotic resistance occurs when bacteria evolve mechanisms to survive exposure to antibiotics.
- Horizontal gene transfer (HGT) is one mechanism by which resistance genes spread between bacterial species.
- A 2022 study by Park and Yamamoto tracked HGT of the mcr-1 gene (which confers resistance to colistin, a last-resort antibiotic) across wastewater samples.
- In hospital wastewater, mcr-1 was detected in 12 of 15 bacterial species sampled.
- In municipal wastewater (downstream from hospitals), mcr-1 was detected in 8 of 15 species sampled.
- The researchers concluded that hospital wastewater is a significant reservoir for the dissemination of colistin resistance genes into the broader environment.

The student wants to present evidence supporting the researchers' conclusion. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Horizontal gene transfer allows resistance genes like mcr-1 to move between different bacterial species, a process that can occur in various aquatic environments.
B) Park and Yamamoto found mcr-1 in a majority of bacterial species in hospital wastewater, and the gene's subsequent detection downstream in municipal wastewater suggests that resistance genes spread from hospitals into the wider environment.
C) Colistin is considered a last-resort antibiotic, and the mcr-1 gene confers resistance to it, making the spread of this gene a significant public health concern.
D) In the 2022 study, 15 bacterial species were sampled in both hospital and municipal wastewater, and mcr-1 was found in at least some species at both sites.

SECTION 2: MATH

MODULE 1 (Same for all students — 22 Questions)

Question 1

Skill: Creating Linear Models from Word Problems | Difficulty: Easy

The width of a rectangle is 6 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

A) 12
B) 46
C) 92
D) 276

Question 2

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

A freelance translator earned a total of \$924 in one week by working at a publishing house and at a law firm. The equation $28h + 18f = 924$ represents this situation, where h is the number of hours worked at the publishing house and f is the number of hours worked at the law firm. Which of the following is the best interpretation of 28 in this context?

A) The amount, in dollars, the translator earned for each hour worked at the publishing house
B) The amount, in dollars, the translator earned for each hour worked at the law firm

C) The number of hours the translator worked in one week at the publishing house
D) The number of hours the translator worked in one week at the law firm

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Question 3

Skill: Linear & Exponential Models from Data | Difficulty: Medium

[Scatterplot: The x-axis ranges from 0 to 8, and the y-axis ranges from 0 to 16. Approximately 12 data points are plotted with an upward trend. A line of best fit passes through approximately (0, 1) and (8, 13). The line has a positive slope of about 1.5 and a y-intercept near 1.]

The scatterplot above shows the relationship between two variables, x and y. A line of best fit is also shown. Which of the following equations best represents the line of best fit shown?

A) $y = 1.0 + 1.5x$
B) $y = 1.0 - 1.5x$
C) $y = -1.0 + 1.5x$
D) $y = -1.0 - 1.5x$

Question 4

Skill: Nonlinear Systems of Equations | Difficulty: Hard

$y = -0.25$ $y = x^2 + 6x + a$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a?

Question 5

Skill: Area & Perimeter | Difficulty: Easy

The width of a rectangle is 5 centimeters. The length of the rectangle is 35 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

A) 10
B) 40
C) 80
D) 200

Question 6

Skill: Ratios, Rates & Proportions | Difficulty: Easy

A jar contains only red and blue marbles. There are 48 red marbles and 32 blue marbles. What fraction of the marbles in the jar are blue?

A) 2/5
B) 3/5
C) 2/3
D) 3/8

Question 7

Skill: Graphs of Linear Equations | Difficulty: Easy

The function g is defined by $g(x) = 5x - 7$. What is the y-intercept of the graph of $y = g(x)$ in the xy-plane?

A) (0, −5)
B) (0, 5)
C) (0, −7)
D) (0, 7)

Question 8

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A biologist observes a colony of bacteria. An exponential model estimates that the population, in thousands, of the colony decreases by 18% every 15.5 hours. Which of the following equations could represent this model, where P is the estimated population, in thousands, t hours after the biologist began observing?

A) P = 50(0.18)^(t + 15.5)
B) P = 50(0.18)^(t/15.5)
C) P = 50(0.82)^(t + 15.5)
D) P = 50(0.82)^(t/15.5)

Question 9

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Medium

The expression $5x^4 + 11x^4 - 3x^4$ is equivalent to bx^4, where b is a constant. What is the value of b?

Question 10

Skill: Radical and Rational Exponents | Difficulty: Medium

Which expression is equivalent to (245y)^(1/2), where y > 1?

A) $245 \cdot \sqrt{y}$
B) $\sqrt{245} \cdot y$
C) $\sqrt{(245y)}$
D) $\sqrt{((245y)^2)}$

Question 11

Skill: Probability - Basic | Difficulty: Medium

At a community fair, there are a total of 350 attendees. Each attendee is in either Hall X, Hall Y, or Hall Z. If one of these attendees is selected at random, the probability of selecting an attendee who is in Hall X is 0.60, and the probability of selecting an attendee who is in Hall Y is 0.28. How many attendees are in Hall Z?

A) 12
B) 42
C) 56
D) 98

Question 12

Skill: Parallel & Perpendicular Lines | Difficulty: Medium

Line m is defined by $y = -(1/3)x + 10$. Line n is perpendicular to line m in the xy-plane. What is the slope of line n?

A) 10
B) 3
C) 1/3
D) 1/10

Question 13

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

$f(x) = -8(3)^{(x/3)}$

Which table gives four values of x and their corresponding values of f(x) for the given exponential function?

A) x: −3, 0, 3, 6 → f(x): −8/3, 0, −24, −48
B) x: −3, 0, 3, 6 → f(x): −8/3, −8, −24, −72
C) x: −3, 0, 3, 6 → f(x): 8/3, −8, −24, −72
D) x: −3, 0, 3, 6 → f(x): −8/3, −8, −24, −48

Question 14

Skill: Function Notation & Evaluation | Difficulty: Medium

The function r is defined by $r(x) = 4x + 7$, and the function q is defined by $q(x) = 5 - x$. What is the value of $3r(2) - q(4)$?

Question 15

Skill: Linear Inequalities | Difficulty: Medium

A school is purchasing laptops for its students. The laptops cost $210 each, and 8% tax is added to the total cost of the order. If the school can spend no more than $68,040 on the laptops, including tax, what is the maximum number of laptops that the school can order?

Question 16

Skill: Radian and Degree Conversion | Difficulty: Medium

The difference between the measure of angle P and the measure of angle Q is $(7/10)\pi$ radians. Which expression shows the difference between the measure of angle P and the measure of angle Q, in degrees?

A) $(7/10) \cdot (180°/\pi)$
B) $(7/10)\pi \cdot (\pi/180°)$
C) $(7/10)\pi \cdot (180°/\pi)$
D) $(7/10)\pi \cdot (360°/\pi)$

Question 17

Skill: Lines & Angles | Difficulty: Hard

In triangle DEF, the measure of angle D is 96°, the measure of angle E is $(2x + 6)°$, and the measure of angle F is $(3x - 12)°$. Point G lies on segment DE, point H lies on segment EF, and segment GH is parallel to segment DF. What is the measure, in degrees, of angle GHE? (Disregard the degree symbol when entering your answer.)

Question 18

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The function f is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at (2, 5) and passes through the points (3, 41) and (0, 149). What is the value of $f(-1) - f(0)$?

A) 108
B) 252
C) 180
D) 288

Question 19

Skill: Linear & Exponential Models from Data | Difficulty: Medium

[Scatterplot: x-axis from 0 to 10, y-axis from 0 to 25. About 10 points with positive trend. Line of best fit passes approximately through (0, 2) and (10, 22), slope ≈ 2.]

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown. Which of the following equations best represents the line of best fit shown?

A) $y = 2 + 2x$
B) $y = 2 - 2x$
C) $y = -2 + 2x$
D) $y = 22 - 2x$

Question 20

Skill: Systems of Linear Equations | Difficulty: Medium

$x + 4 = 3y + 10$ $x - 4 = -3y + 2$

The solution to the given system of equations is (x, y). What is the value of $3x$?

A) $-15/7$
B) 9
C) 18
D) 36

Question 21

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

A chemist monitors a radioactive isotope. An exponential model estimates that the mass, in milligrams, of the sample decreases by 30% every 18.4 hours. Which of the following equations could represent this model, where M is the estimated mass, in milligrams, of the sample t hours after the chemist began monitoring?

A) M = 200(0.30)^(t + 18.4)
B) M = 200(0.30)^(t/18.4)
C) M = 200(0.70)^(t + 18.4)
D) M = 200(0.70)^(t/18.4)

Question 22

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The function g is a quadratic function. In the xy-plane, the graph of $y = g(x)$ has a vertex at (3, 10) and passes through the points (4, 46) and (1, 154). What is the value of $g(-2) - g(0)$?

A) 432
B) 576
C) 720
D) 864

MODULE 2 — EASIER PATH (22 Questions)

(For students scoring below ~60% on Module 1)

Question 1

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

In one month in 2020, a dog walker earned a total of $840 by walking dogs in a neighborhood and at a park. The equation $15n + 12p = 840$ represents this situation, where n is the number of walks in the neighborhood and p is the number of walks at the park. Which of the following is the best interpretation of 15 in this context?

A) The amount, in dollars, the dog walker earned per neighborhood walk
B) The amount, in dollars, the dog walker earned per park walk
C) The number of walks the dog walker completed in the neighborhood
D) The number of walks the dog walker completed at the park

Question 2

Skill: Creating Linear Models from Word Problems | Difficulty: Easy

A veterinarian uses a linear model to estimate the weight of a growing puppy after it is born. The model estimates that a certain puppy weighs 2,400 grams at birth and gains 85 grams per day for 90 days after it is born. Based on this model, what is the estimated weight, in grams, of this puppy 10 days after it is born?

Question 3

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

$m - 27n = p$

The given equation relates the positive numbers m, n, and p. Which equation correctly expresses m in terms of n and p?

A) $m = p - 27n$
B) $m = p + 27n$
C) $m = 27np$
D) $m = p/(27n)$

Question 4

Skill: Linear Inequalities | Difficulty: Medium

[Graph: xy-plane with a solid line passing through approximately (0, 7) and (2, 0), with the region below and to the left of the line shaded. The line has a negative slope of approximately −3.5.]

The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

A) (−3, 0)
B) (0, 8)
C) (0, −5)
D) (5, 0)

Question 5

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Easy

A rectangle has a length that is 36 times its width. The function $y = (36w)(w)$ represents this situation, where y is the area, in square meters, of the rectangle and $y > 0$. Which of the following is the best interpretation of 36w in this context?

A) The length of the rectangle, in meters
B) The area of the rectangle, in square meters
C) The difference between the length and the width of the rectangle, in meters
D) The width of the rectangle, in meters

Question 6

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $7(x + 3) = 6(x + 3) + 28$, what is the value of $x + 3$?

A) -4
B) 25
C) 28
D) 31

Question 7

Skill: Systems of Linear Equations | Difficulty: Medium

$x + 8 = -5y + 3$ $x - 8 = 5y + 19$

The solution to the given system of equations is (x, y). What is the value of $2x$?

A) $-7/5$
B) 11
C) 22
D) 44

Question 8

Skill: Creating Linear Models from Word Problems | Difficulty: Easy

In 10 days, a squirrel gathered 52.5 acorns. Which equation describes the number of acorns y the squirrel gathered in these 10 days?

A) $y = 10$
B) $y = 52.5/10$
C) $y = 52.5 + 10$
D) $y = 52.5$

Question 9

Skill: Volume & Surface Area | Difficulty: Medium

To study porosity in building materials, samples of concrete were cut in the shape of a cube. The length of the edge of one of these cubes is 4.000 centimeters. This cube has a density of 0.350 grams per cubic centimeter. What is the mass of this cube, in grams?

Question 10

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Medium

$-3|7x + 4| + 6 = -6$

What are all solutions to the given equation?

A) 0
B) 0 and −8/7
C) 0 and −15/7
D) There is no solution.

Question 11

Skill: Graphs of Linear Equations | Difficulty: Medium

For the linear function h, h(d) = −4, where d is a constant, h(3) = 32, and the slope of the graph of y = h(x) in the xy-plane is 9. For the linear function u, u(d) = −4 and u(6) = 52. What is the slope of the graph of y = u(x) in the xy-plane?

A) −2
B) 7
C) 8
D) 9

Question 12

Skill: Linear Equations in One & Two Variables | Difficulty: Medium

[Graph: xy-plane from −10 to 10 on both axes. A line passes through approximately (−3, −10) and (0, 8), with a positive slope of roughly 6. The line crosses the y-axis at about y = 8.]

The graph shows a linear relationship between x and y. Which equation represents this relationship, where R is a positive constant?

A) $Rx + 6y = 48$
B) $Rx - 6y = -48$
C) $6x + Ry = 48$
D) $6x - Ry = -48$

Question 13

Skill: Linear Inequalities | Difficulty: Hard

[Graph: xy-plane. A dashed line passes through (−9, 0) and (0, −12), with the region above and to the right of the line shaded. The line has a negative slope.]

The shaded region shown represents the solutions to $rx + ty \geq -108$, where r and t are constants. What is the value of $r + t$?

A) −3
B) 3
C) −21
D) 21

Question 14

Skill: Volume & Surface Area | Difficulty: Hard

The edge length, in inches, of cube B is 5/72 the edge length, in inches, of cube A. The surface area, in square inches, of cube B is n times the surface area, in square inches, of cube A. What is the value of n?

A) 125/373248
B) 5/432
C) 25/5184
D) 25/72

Question 15

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[Graph: xy-plane showing a decreasing curve. As x moves left (decreasing), the curve approaches y = 4 from below. The curve passes through (0, 3), then drops steeply as x increases, passing through (0.5, approximately 2) and continuing downward.]

The graph of $y = g(x) + 3$ is shown. Which equation defines function g?

A) $g(x) = -4^x + 1$
B) $g(x) = -4^x + 3$
C) $g(x) = -4^x + 4$
D) $g(x) = -4^x + 5$

Question 16

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

Which expression is equivalent to $12(x + 8)$?

A) $12x + 8$
B) $12x + 96$
C) $12x + 20$
D) $12x + 12$

Question 17

Skill: Ratios, Rates & Proportions | Difficulty: Medium

A square blueprint has a side length of 40 inches, and 1 inch on the blueprint represents an actual distance of 16 miles. A smaller version of the same blueprint is printed as a square with the side length 60% shorter than the side length of the previous blueprint. On the smaller blueprint, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 6.40
B) 9.60
C) 25.60
D) 40.00

Question 18

Skill: Systems of Linear Equations | Difficulty: Hard

$y = x - d$ $y = -(x - 3)^2$

In the given system of equations, d is a positive constant. The system has two distinct real solutions. Which of the following could be the value of d?

A) 1
B) 2
C) 11/4
D) 4

Question 19

Skill: Similar Triangles | Difficulty: Hard

In triangle GHI, the measure of angle G is 72° and GI = 25. In triangle JKL, the measure of angle J is 72° and JL = 100. Which additional piece of information is sufficient to prove that triangle GHI is similar to triangle JKL?

A) GH = 30 and JK = 30.
B) GH = 30 and KL = 120.
C) The measures of angle H and angle L are 40° and 68°, respectively.
D) The measures of angle H and angle K are 72° and 50°, respectively.

Question 20

Skill: Circle Equations | Difficulty: Hard

The perimeter of an equilateral triangle is 720 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?

Question 21

Skill: Function Notation & Evaluation | Difficulty: Hard

The function g is defined by $g(x) = |x|/a + 18$, where $a < 0$. What is the product of $g(20a)$ and $g(9a)$?

Question 22

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

The function h is defined by h(x) = ab^(x/n), where a, b, and n are constants, and b and n are integers. If h(4) = 9 and h(7) = 243, what is the value of h(9)?

MODULE 2 — HARDER PATH (22 Questions)

(For students scoring above ~60% on Module 1)

Question 1

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

A caterer charges a flat fee plus a per-person cost for events. The equation C = 250 + 18g represents the total cost C, in dollars, for an event with g guests. Which of the following is the best interpretation of 250 in this context?

A) The flat fee, in dollars, charged regardless of the number of guests
B) The cost per guest
C) The total cost for 250 guests
D) The number of guests at which the total cost equals $250

Question 2

Skill: Creating Linear Models from Word Problems | Difficulty: Medium

An oceanographer uses a linear model to estimate the depth of a submarine after it begins descending. The model estimates that the submarine is at a depth of 0 meters at the start and descends at a rate of 12.5 meters per minute for 80 minutes. Based on this model, what is the estimated depth, in meters, of the submarine 6 minutes after it begins descending?

Question 3

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

$p - 42r = s$

The given equation relates the positive numbers p, r, and s. Which equation correctly expresses p in terms of r and s?

A) $p = s + 42r$
B) $p = s - 42r$
C) $p = 42rs$
D) $p = s/(42r)$

Question 4

Skill: Systems of Linear Equations | Difficulty: Hard

$y = x - k$ $y = -2(x - 5)^2$

In the given system of equations, k is a positive constant. The system has two distinct real solutions. Which of the following could be the value of k?

A) 3
B) 4
C) 39/8
D) 6

Question 5

Skill: Linear Inequalities | Difficulty: Medium

A company is purchasing office chairs for its employees. The chairs cost $185 each, and 6% tax is added to the total cost of the order. If the company can spend no more than $49,210 on the chairs, including tax, what is the maximum number of chairs the company can purchase?

Question 6

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Hard

$-2|9x + 6| + 8 = -4$

What are all solutions to the given equation?

A) 0
B) 0 and −2/3
C) 0 and −12/9
D) There is no solution.

Question 7

Skill: Graphs of Linear Equations | Difficulty: Hard

For the linear function v, v(c) = −1, where c is a constant, v(2) = 23, and the slope of the graph of y = v(x) in the xy-plane is 8. For the linear function w, w(c) = −1 and w(7) = 71. What is the slope of the graph of y = w(x) in the xy-plane?

A) −2
B) 8
C) 9
D) 12

Question 8

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[Graph: A curve in the xy-plane. As x moves left (decreasing), the curve approaches y = 9 from below. The curve passes through (0, 8) and (1, 4), then drops steeply below zero as x increases.]

The graph of y = f(x) + 2 is shown. Which equation defines function f?

A) f(x) = −5^x + 1
B) f(x) = −5^x + 5
C) f(x) = −5^x + 9
D) f(x) = −5^x + 7

Question 9

Skill: Volume & Surface Area | Difficulty: Hard

The edge length, in centimeters, of cube Q is 7/90 the edge length, in centimeters, of cube P. The surface area, in square centimeters, of cube Q is n times the surface area, in square centimeters, of cube P. What is the value of n?

A) 49/8100
B) 343/729000
C) 7/540
D) 49/90

Question 10

Skill: Lines & Angles | Difficulty: Hard

In triangle MNO, the measure of angle M is $(4y - 5)°$, the measure of angle N is $y°$, and the measure of angle O is $(3y - 23)°$. Point P lies on segment MN, point Q lies on segment NO, and segment PQ is parallel to segment MO. What is the measure, in degrees, of angle PQN? (Disregard the degree symbol when entering your answer.)

Question 11

Skill: Systems of Linear Equations | Difficulty: Medium

$2x + 6 = -3y + 1$ $2x - 6 = 3y + 27$

The solution to the given system of equations is (x, y). What is the value of $4x$?

A) 28
B) 14
C) −12/3
D) 56

Question 12

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The function p is a quadratic function. In the xy-plane, the graph of $y = p(x)$ has a vertex at $(-1, 3)$ and passes through the points $(0, 11)$ and $(-3, 35)$. What is the value of $p(2) - p(-1)$?

A) 69
B) 72

C) 75
D) 78

Question 13

Skill: Circle Equations | Difficulty: Hard

The perimeter of an equilateral triangle is 594 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?

Question 14

Skill: Function Notation & Evaluation | Difficulty: Hard

The function f is defined by $f(x) = |x|/b + 10$, where $b < 0$. What is the product of $f(16b)$ and $f(4b)$?

Question 15

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

The function f is defined by $f(x) = cd^{(x/m)}$, where c, d, and m are constants, and d and m are integers. If $f(3) = 16$ and $f(6) = 256$, what is the value of $f(9)$?

Question 16

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

Which expression is equivalent to $18(x + 7)$?

A) $18x + 7$
B) $18x + 126$
C) $18x + 25$
D) $18x + 18$

Question 17

Skill: Ratios, Rates & Proportions | Difficulty: Hard

A square map has a side length of 50 inches, and 1 inch on the map represents an actual distance of 20 miles. A smaller version of the same map is printed as a square with the side length 80% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 4.00
B) 16.00
C) 80.00
D) 100.00

Question 18

Skill: Similar Triangles | Difficulty: Hard

In triangle ABC, the measure of angle A is 65° and AB = 40. In triangle DEF, the measure of angle D is 65° and DE = 120. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle DEF?

A) AC = 45 and DF = 45.
B) AC = 45 and EF = 135.
C) The measures of angle B and angle F are 50° and 65°, respectively.
D) The measures of angle B and angle E are 65° and 40°, respectively.

Question 19

Skill: Probability - Basic | Difficulty: Medium

A bag contains 150 tokens. Each token is either gold, silver, or bronze. If one token is selected at random, the probability it is gold is 0.52 and the probability it is silver is 0.30. How many tokens are bronze?

A) 18
B) 27
C) 45
D) 78

Question 20

Skill: Parallel & Perpendicular Lines | Difficulty: Medium

Line j is defined by $y = -(1/5)x + 9$. Line k is perpendicular to line j in the xy-plane. What is the slope of line k?

A) 9
B) 5
C) 1/5
D) 1/9

Question 21

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

The expression $9x^3 + 4x^3 - 6x^3$ is equivalent to bx^3, where b is a constant. What is the value of b?

Question 22

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The function q is a quadratic function. In the xy-plane, the graph of $y = q(x)$ has a vertex at $(-2, 4)$ and passes through $(-1, 13)$ and $(0, 40)$. What is the value of $q(1) - q(-2)$?

A) 72
B) 77
C) 81
D) 85